

Practical No. 12

ON

SSCA4030 - Image Processing

TITLE: Morphological Operations in Image Processing

**BACHLEOR OF SCIENCE IN INFORMATION TECHNOLOGY
(HONOURS.)**

SUBMITTED BY:

Name: Shah Rishabh Alpeshbhai

Enrolment: 23SS02IT181

Class: BSCIT-H – 7A - Batch 2023-27

```
import cv2
import matplotlib.pyplot as plt

# Read image
img = cv2.imread("/image-removebg-preview (4).png", 0)

# Create binary image
_, binary = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)

# Create structuring element
kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (3, 3))

# Morphological operations
erosion = cv2.erode(binary, kernel, iterations=1)
dilation = cv2.dilate(binary, kernel, iterations=1)
opening = cv2.morphologyEx(binary, cv2.MORPH_OPEN, kernel)
closing = cv2.morphologyEx(binary, cv2.MORPH_CLOSE, kernel)

# Display results
plt.figure(figsize=(10, 8))

<Figure size 1000x800 with 0 Axes>
<Figure size 1000x800 with 0 Axes>

plt.subplot(2, 3, 1)
plt.imshow(binary, cmap='gray')
plt.title("Original")
plt.axis("off")

plt.subplot(2, 3, 2)
plt.imshow(erosion, cmap='gray')
plt.title("Erosion")
plt.axis("off")

plt.subplot(2, 3, 3)
plt.imshow(dilation, cmap='gray')
plt.title("Dilation")
plt.axis("off")

plt.subplot(2, 3, 5)
plt.imshow(opening, cmap='gray')
plt.title("Opening")
plt.axis("off")

plt.subplot(2, 3, 6)
plt.imshow(closing, cmap='gray')
plt.title("Closing")
plt.axis("off")

plt.show()
```

